

Shuvom Sadhuka

Stata Center G574 • ssadhuka@mit.edu • shuvom-s.github.io

EDUCATION

Massachusetts Institute of Technology	Cambridge, MA
PhD candidate in Computer Science	2027 (expected)
Research Interests: Reliable and trustworthy machine learning, AI evaluation, uncertainty quantification, applications to biomedical domains	
Massachusetts Institute of Technology	Cambridge, MA
SM in Computer Science	September 2023
Concentration: AI	
Harvard University	Cambridge, MA
AB in Computer Science and Statistics	May 2022

SELECTED HONORS & AWARDS

MIT Envisioning the Future of Computing Essay, Honorable Mention	2023
National Science Foundation Graduate Research Fellowship	2022
Hertz Foundation Fellowship	2022

PREPRINTS & PUBLICATIONS (* co-first)

[w] = working paper and/or under review, [p] = full publication

[w] S. Sadhuka, D. Prinster, C. Fannjiang, G. Scalia, A. Regev, H. Wang. *E-evaluator: Reliable Agent Verifiers with Sequential Hypothesis Testing*. Preprint 2026 ([arxiv](https://arxiv.org/abs/2601.00000))

[w] S. Sadhuka, S. Dandapanthula, S. Khan, M. Oberst, B. Berger, A. Chouldechova. *Calibrated but Correlated: Shared Residuals among LLM Judges Distorts Model Selection*. In progress.

[p] S. Balachandar, S. Sadhuka, B. Berger, E. Pierson, N. Garg. *Urban Incident Prediction with Graph Neural Networks: Integrating Government Ratings and Crowdsourced Reports*. AAAI 2026

[p] D. Shanmugam*, S. Sadhuka*, M. Raghavan, J. Guttag, B. Berger, E. Pierson. *Evaluating multiple models using labeled and unlabeled data*. NeurIPS 2025

[p] S. Sadhuka, S. Lin, B. Berger**, E. Pierson**. *A Bayesian Model for Multi-stage Censoring*. ML4H 2025 (proceedings track), also **spotlight presentation at ML4H 2024** (findings track)

[p] H. Cho, D. Froelicher*, N. Dokmai*, A. Nandi*, S. Sadhuka*, M. Hong*, B. Berger. *Privacy-Enhancing Technologies in Biomedical Data Science*. *Annual Reviews in Biomedical Data Science* 2024

[p] S. Sadhuka, D. Fridman, B. Berger, H. Cho. *Assessing transcriptomic reidentification risks using discriminative sequence models*. *Genome Research* 2023; **oral presentation at RECOMB 2023**

[p] H. Pirie, S. Sadhuka, J. Wang, R. Andrei, J. Hoffman. *Topological phononic logic*. **Cover article in** *Physical Review Letters* 2022

- Press: [Science Daily](#), [Harvard SEAS](#), [IEEE Spectrum](#), [Hackaday](#)

[p] Q. Wang, D. Kelley, J. Ulirsch, M. Kanai, S. Sadhuka, R. Cui, C. Albors, N. Cheng, Y. Okada, Biobank Japan Project, F. Aguet, K. Ardlie, D. MacArthur, H. Finucane. *Leveraging supervised learning for functionally informed fine-mapping of cis-eQTLs identifies an additional 20,913 putative causal eQTLs*. *Nature Communications* 2021

TALKS & PRESENTATIONS

E-evaluator: Reliable Agent Verifiers with Sequential Hypothesis Testing

Rational Intelligence Seminar Series [[YouTube](#)] 2026

MIT, Algorithms Coffee 2026

Evaluating multiple models using labeled and unlabeled data

MIT, Machine Learning Tea 2025

Assessing transcriptomic reidentification risks

RECOMB, Oral Presentation [[YouTube](#)] 2024

TEACHING EXPERIENCE

Massachusetts Institute of Technology

Cambridge, MA

TA, 18.418: *Topics in Computational Molecular Biology* (Prof. Bonnie Berger) Fall 2023

Harvard University

Cambridge, MA

TF, CS 124: *Data Structures and Algorithms* (Prof. Michael Mitzenmacher) Spring 2021, 2022

TF, MCB 112: *Biological Data Analysis* (Prof. Sean Eddy) Fall 2020

TF, Stat 110: *Introduction to Probability* (Prof. Joe Blitzstein) Fall 2019

REVIEWING AND SERVICE

Roundtable Chair, ML4H 2024, 2025 2024, 2025

Member, NIST AI Safety Institute Task Force on Red Teaming 2024

Planning Committee, Hertz Foundation Summer Workshop 2023

Reviewing: NeurIPS, ICML, ICLR, Journal of Computational Biology, *PNAS*

LEADERSHIP & OUTREACH

Mentor Training Chair, MIT EECS Graduate Application Assistance Program. 2024-present

- Mentor, 2022-present

Tutor, Research Science Institute Summer 2023, 2024

Non-Resident Tutor, Mather House (Harvard University) 2022-2023

Co-captain, Harvard College Bhangra 2020-2022

MENTORSHIP

Ragulan Sivakumar (MEng student) Fall 2024-Spring 2025

Sophia Lin (RSI summer student) Summer 2024-Fall 2024

INDUSTRIAL EXPERIENCE

Abridge AI New York, NY

PhD Research Intern 2026

Mentors: Alex Chouldechova, Michael Oberst

Genentech San Francisco, CA

ML Research Intern 2025

Mentors: Hanchen Wang, Aviv Regev

BBN Technologies Cambridge, MA

Software Intern, Machine Translation Division 2020

REFERENCES

Bonnie Berger
Massachusetts Institute of Technology
Simons Professor of Mathematics and of Computer Science
bab@csail.mit.edu

Hanchen Wang
Google DeepMind
Research Scientist
hanchenw@cs.stanford.edu

Emma Pierson
University of California, Berkeley
Assistant Professor of Electrical Engineering and Computer Science
emma.pierson@berkeley.edu